E-COMMERCE SALES ANALYSIS PROJECT

- WHERE EVERY TRANSACTION TELLS A STORY

ABOUT PROJECT

Dataset

9,994 Orders

Tool (Python)

Pandas + Plotly

Tool (SQL)

MySQL 8.0

Domain

Superstore Sales

Analyze e-commerce sales using Python & SQL: Discover total revenue, top-performing categories, customer segment behaviour, monthly trends, and profit margins — enabling smarter, data-driven business decisions.

DATASET OVERVIEW & DATA CLEANING

```python
data = pd.read_csv("Sample - Superstore.csv",
           encoding="latin1")

data['Order Date'] = pd.to_datetime(
    data['Order Date'])

print(data.isnull().sum())
```

Result Grid

Column	Type
Order Date	datetime64
Ship Date	datetime64
Sales	float64
Null Values	0 (Clean)

- 9,994 total records
- 21 columns
- No null values
- 4 years of data

TOTAL REVENUE & PROFIT OVERVIEW

```sql
SELECT
  ROUND((SUM(profit) / SUM(sales))
    * 100, 2)
  AS profit_margin_percentage
FROM orders;
```

```python
total_sales = data['Sales'].sum()
total_profit = data['Profit'].sum()
```

TOTAL REVENUE

$2,297,201

Gross Sales

TOTAL PROFIT

$286,397

Net Profit

PROFIT MARGIN

12.47%

Overall Margin

SALES ANALYSIS BY CATEGORY

```
sales_by_category = data.groupby(
  'Category')['Sales'].
  sum().reset_index()

fig = px.pie(sales_by_category,

  values='Sales',  names='Category')
```

Result Grid

Category	Sales ($)
Technology	836,154
Furniture	741,999
Office Supplies	719,047

Revenue Breakdown

Technology

36.4% $836K

Furniture

32.3% $742K

Office Supplies

31.3% $719K

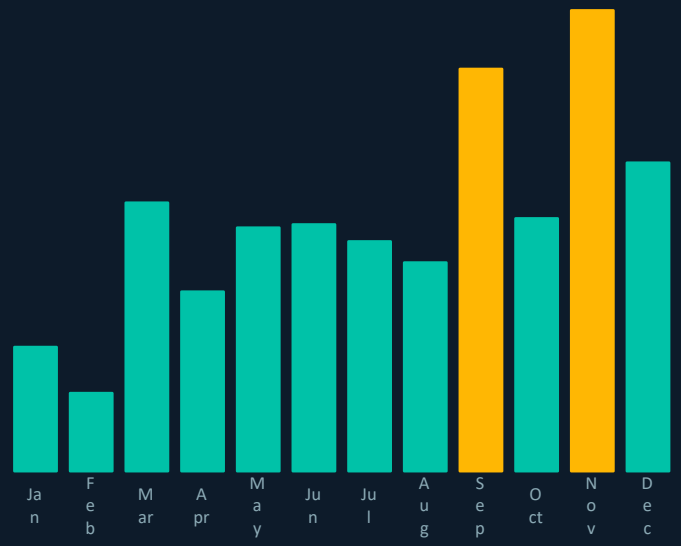
MONTHLY SALES ANALYSIS

```
sales_by_month = data.groupby(
  'Order Month')['Sales'].
  sum().reset_index()

fig = px.line(sales_by_month,

  x='Order Month',  y='Sales',

  title='Monthly Sales Analysis')
```

Result Grid (selected months)

Month	Total Sales ($)
September (9)	307,649
November (11)	352,461
December (12)	325,294

Monthly Trend



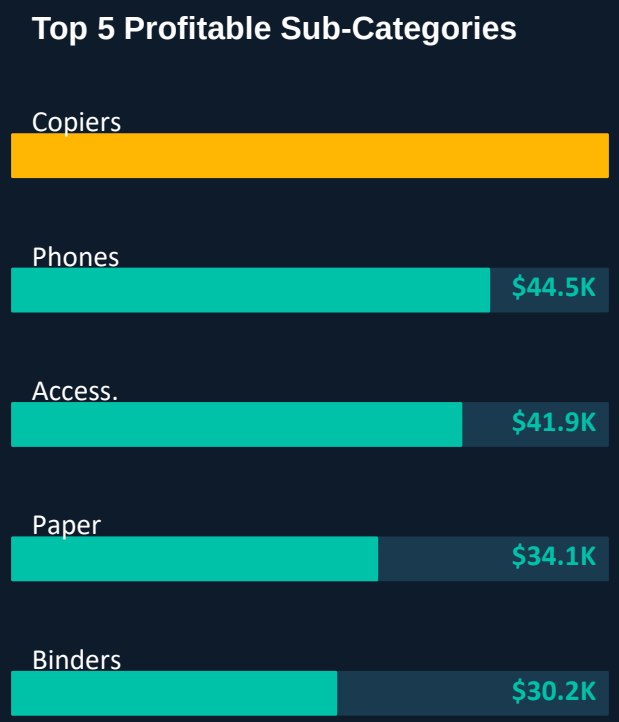
PROFIT ANALYSIS BY SUB-CATEGORY

```
Profit_by_sub = data.groupby(
  'Sub-Category')['Profit'].

  sum().reset_index()

fig = px.bar(Profit_by_sub,

  x='Sub-Category',

  y='Profit',

  title='Profit By Sub Category')
```

Result Grid

Sub-Category	Profit ($)
Copiers	55,618
Phones	44,516
Accessories	41,937
Paper	34,054

⚠ Tables (-17,725), Bookcases (-3,473), Supplies (-1,189) = LOSSES



SALES VS PROFIT BY CUSTOMER SEGMENT

```
seg_data = data.groupby('Segment').
  agg({'Sales':'sum',
       'Profit':'sum'}).reset_index()
seg_data['ratio'] =
  seg_data['Sales'] /
  seg_data['Profit']
```

Result Grid

Segment	Sales ($)	Ratio
Consumer	1,161,401	8.66
Corporate	706,146	7.68
Home Office	429,653	7.13

Segment Breakdown

Consumer

Sales: $1.16M

Profit: $134K

50.6 %

Corporate

Sales: $706K

Profit: $92K

30.7 %

Home Office

Sales: $430K

Profit: $60K

18.7 %

YEAR-OVER-YEAR SALES GROWTH

```
SELECT year, total_sales,
LAG(total_sales) OVER
  (ORDER BY year)
  AS prev_year_sales,
ROUND((total_sales -
  LAG(total_sales) OVER
  (ORDER BY year)) /
  LAG(total_sales) OVER
  (ORDER BY year)) * 100, 2)
  AS growth_percentage
```

Result Grid

Year	Total Sales	Growth %
2014	$484,247	—
2016	$609,206	+29.46%
2017	$733,215	+20.36%

YoY Growth Results

2014
Sales: $484K
—

2015
Sales: $470K
-2.83%

2016
Sales: $609K
+29.46%

2017
Sales: $733K
+20.36%

THANK YOU

FOR ATTENTION

2026 E-COMMERCE SALES PRESENTATION